

Lab 4: VibeCoding — Chat History

Marble Tilt Maze · PES1UG24CS706 Aadarsh · Tool: Claude (Claude Code, desktop app) · 30 Sep 2026
Grey monospace lines are actions Claude took (running commands, reading and editing files).

ME (AADARSH)

@"/Users/aadarsh/Downloads/Lab_4_VibeCoding_Student_handout (1).pdf" go through this

- Bash: Extract text from the lab handout PDF
- Read: /Users/aadarsh/Downloads/Lab_4_VibeCoding_Student_handout (1).pdf
- Bash: Install pypdf in scratch venv and extract PDF text

CLAUDE

I read the handout. It's a single page, and here's what it asks for.

Lab 4: VibeCoding (60 minutes, individual work)

Goal: Use an AI coding tool (ChatGPT, Gemini, Claude, etc.) to fix a broken Python game or puzzle and add the features listed in its README. The handout wants you to get each task done in about 3–4 prompts.

Steps:

1. Go to the repo you've been assigned.
2. Read its README to see what you need to deliver.
3. Clone or fork the repo.
4. Run the Python game or puzzle.
5. **Record about 10 seconds of video before changing anything.**
6. Prompt the AI to fix the broken code.
7. Prompt it to add each feature from the README, one at a time.
8. **Record another 10 seconds of video after all the changes.**
9. Push the code to your own repo on your personal account.
10. **Don't open a pull request to the main repo (SETAPESU26).**

What to upload, in a Lab-4 folder in your Lab repo:

- The before and after videos
- The updated code
- Your chat history, exported as a doc or PDF

Things to keep in mind:

- You'll need to export the chat history, so fix the bugs and add the features one at a time in separate, focused prompts. The graders are looking at how well you prompt.
- Record the "before" video before you touch any code.
- Your fork will have the original repo as its upstream, so double-check that you're pushing to your own repo and not opening a PR to SETAPESU26.

If you'd like to do the lab here, send me the assigned repo's URL. I'll clone it into a separate folder, not this Monolith project, and we can go through the README together. You'll need to record the videos yourself.

ME (AADARSH)

i have cloned it and as you have access to my github just go through the repo and check if everything is done properly

- Bash: Find recently cloned repos and list GitHub repos
- Bash: Locate local clones and inspect both GitHub repos
- Bash: List files in Lab repo, fork, and search for local clone
- Bash: Search for local clone and read the fork's README

CLAUDE

I checked your GitHub and your machine. Almost none of Lab 4 has been done yet.

Your fork, [HarryGJackson/42_marble-tilt-maze](#)

- It has only one commit, the original "Initial Marble Tilt Maze starter" from 24 Sep. None of your changes are on GitHub.
- It also contains committed `__pycache__/*.pyc` files. It would be tidy to remove those and add a `.gitignore`.

Your local clone

- I couldn't find one. There's no `marble.py` anywhere in your home folder. You may have cloned it somewhere I didn't search, or the clone didn't finish. Tell me the path if you have it.

Your Lab repo, [PES1UG24CS706_Aadarsh_Lab_SE](#)

- It has `lab 1`, `lab 2` and `lab 3`, but no `Lab-4` folder yet. Your earlier folders are named like `lab 3`, so `lab 4` would match them better than `Lab-4`.

What's still left, from the handout and the repo's README:

Item	Status
Task 1: fix the marble bouncing off empty space near wall corners	Not done
Task 2: end screen showing win with finish time or time-up, then waiting for input	Not done
Task 3: replay with Easy, Medium or Hard difficulty (tilt strength, friction, time limit), or exit	Not done
Task 4: sounds for bouncing off a wall, reaching the goal and running out of time	Not done
10-second video before your changes	You need to record it
10-second video after your changes	You need to record it
Chat history exported as a doc or PDF (the README asks for the chat link instead)	Not done
Updated code, videos and chat in your Lab repo's Lab 4 folder	Not done

Record the "before" video first, while the code is still the original starter.

I can clone the fork, run the game, and work through the four tasks one prompt at a time so the chat history stays clean to export. Then I'd push to your fork and to the Lab 4 folder. I won't open a PR to SETAPESU26. Do you want me to go ahead? Keep in mind the handout says it's an individual assignment and your prompts are part of what's graded, so you may want to write the prompts yourself.

ME (AADARSH)

oh no, i cloned it into a completely different repo

- Bash: Check Monolith status and search for marble files
- Bash: List all local git repos and their remotes

- Bash: Inspect the Sem5/SE repo
- Bash: Find nested git repo inside SE lab folders
- Bash: Search Spotlight and other locations for the maze code

CLAUDE

I couldn't find the maze code anywhere on your Mac, so it's not something that needs untangling. I checked every git repo under your home folder, ran a Spotlight search for `game_engine.py`, and looked in `/tmp`, iCloud Drive and external drives.

- **Monolith** (the internship repo we're in) is clean. Nothing was cloned into it.
- **Documents/Sem5/SE/Lab 1** is your local clone of the Lab_SE repo. It has only its usual files plus three untracked ones: `.DS_Store` and two `.pages` files. No maze code.
- Your other repos (pes-vcs, wedding-mua-autopilot, nvm) don't have it either.

Most likely the clone failed, or it went into a folder you've since deleted.

Where do you think you cloned it? If you don't remember, I can clone your fork fresh into `Documents/Sem5/SE/Lab 4`, next to your other lab folders, and we'll start from there.

ME (AADARSH)

https://github.com/HarryGJackson/42_marble-tilt-maze

- Bash: Clone the fork into a Lab 4 folder and inspect it
- Bash: Read the game source and check pygame install

CLAUDE

Your fork is cloned to `Documents/Sem5/SE/Lab 4/42_marble-tilt-maze`, next to your other lab folders and outside Monolith. It's the untouched starter code, one commit, and pygame 2.6.1 is already installed, so it should run straight away.

The four README tasks are all still to do. Here's where each one lives in the code:

Task	Current state	Where
1. Collision detection	It checks the marble's square bounding box against the wall. Near corners the box's edge hits the wall before the round marble does, so it bounces off empty space. The fix is a proper circle-vs-rectangle check.	<code>game/game_engine.py</code> , <code>_resolve_wall_collisions</code>
2. Game over screen	Win or time-up is only printed to the terminal, and the game freezes on the last frame.	<code>game/game_engine.py</code> , end of <code>render</code>
3. Replay and difficulty	Doesn't exist. Tilt strength (0.6), friction (0.02) and the time limit (45s) are fixed values in <code>__init__</code> .	<code>game/game_engine.py</code> and <code>main.py</code>
4. Sounds	Doesn't exist. There are no sound files, so the sounds will need to be generated in code or added as files.	New

Two small things:

- The repo includes committed `__pycache__/*.pyc` files. You could remove them and add a `.gitignore`.
- The README asks for the chat link, but the handout asks for the chat history exported as a doc or PDF. Doing both is safest.

Next step: record the "before" video now, while the code is untouched:

```
cd "/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze" && python3 main.py
```

To show the Task 1 bug in the video, steer the marble along the underside of the top wall towards its right end (around $x=460$, $y=140$). You'll see it bounce off empty space near the corner.

Once you've recorded it, tell me whether you want to write the prompts yourself (they're part of the grade) or have me do the tasks one at a time.

ME (AADARSH)

i don't have anything on my laptop

- Bash: Confirm clone exists, show machine name, open folder in Finder

CLAUDE

The code is on your laptop now. I cloned it a moment ago, and this session is running on your MacBook Pro (Aadarshs-Macbook-Pro.local). I've opened the folder in Finder for you. It's at:

Documents > Sem5 > SE > Lab 4 > 42_marble-tilt-maze

Inside it are README.md, main.py, requirements.txt and the game/ folder. Python and pygame are already installed, so you can run it now:

```
cd "/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze" && python3 main.py
```

To record the "before" video, press **Cmd + Shift + 5**, choose "Record Selected Portion" or "Record Entire Screen", and play for about 10 seconds.

If you meant something else, like no screen recorder or no editor, tell me what's missing.

ME (AADARSH)

can i run on idle i prefer idle

- Bash: Check which IDLE and Python are installed
- Bash: Check Homebrew Python for pygame and open main.py in IDLE

CLAUDE

Yes, you can use IDLE. I've opened main.py in it. To run the game, press **F5** (or choose Run > Run Module) and the game window will open.

A few things to know:

- **Use the IDLE app in Applications > Python 3.14**, which is the one I opened. That Python has pygame installed. You also have a Homebrew Python whose IDLE doesn't have pygame, so it would fail with No module named 'pygame'.
- **The game window may open behind IDLE.** If nothing seems to happen after F5, check the Dock or press Cmd + Tab.
- **Always run main.py itself**, not the files inside game/. Those files can't be run on their own.
- **The win and time-up messages appear in the IDLE Shell window.** Right now that's the only place the game shows them, which is the Task 2 bug. Close the game window to stop it.

ME (AADARSH)

@"/Users/aadarsh/Library/Group

Containers/group.com.apple.screencapture/ScreenRecordings/5F076C4A-A77C-49AA-88FA-

D2F41F285035.mov" is this fine for the before ?

- Bash: Check the recording's size, duration and available video tools
- Bash: Get video duration and extract sample frames with AVFoundation
- Bash: Copy the recording to scratchpad and extract frames
- Bash: Look for where the recording was saved
- Bash: Extract duration and sample frames from the saved recording

CLAUDE

Still checking the recording: the first tool couldn't read it, so I'm trying another way.

- Bash: Read recording metadata and copy it to scratchpad
- Bash: Extract sample frames from the recording copy
- Bash: Search for today's screen recording anywhere
- Bash: Copy recording via glob and extract frames
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/f0.jpg
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/f2.jpg
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/f3.jpg
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/f5.jpg
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/f4.jpg

CLAUDE

Yes, it works as your "before" video. I pulled frames from it, and it shows the original game running from IDLE with the timer counting down (38s to 26s) and the marble moving through the maze. Nothing has been changed yet, which is what matters.

A few things you could improve if you have time:

- **It's 23 seconds long, and the handout asks for 10.** You can trim it in QuickTime with Edit › Trim (⌘T), or just leave it; the extra length probably won't matter.
- **It never reaches the end of a game.** In the last 8 seconds the marble sits in the bottom-right corner next to the green goal without touching it. The "before" video is meant to show what's broken, so a better recording would:
 1. Roll the marble into the goal, or wait for the timer to run out.
 2. Show that the game just freezes and the only result is a line printed in the IDLE Shell (Solved! Finished in ...s), not an end screen. That's the Task 2 bug.
 3. Optionally, drag the marble along the underside of the top wall towards its right end, where it bounces before touching the wall. That's the Task 1 bug.
- **The file name has a hidden special character before "AM"**, which can make it awkward to work with. Rename it to something like before.mov when you move it into your Lab 4 folder.

It's about 33 MB, which is fine to push to GitHub.

When you're ready, tell me whether you'll write the prompts for the four tasks yourself or want me to do them one at a time.

ME (AADARSH)

ill upload this video now lets go to task 1

- Edit: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py

- Read: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py
- Edit: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py
- Read: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py

CLAUDE

Now a quick headless test comparing the old and new behaviour near a corner:

- Bash: Test new circle-vs-rect collision headlessly

CLAUDE

Task 1 is fixed. The marble now only bounces when it actually touches a wall, including near corners.

The cause: the game treated the round marble as a square box. Near a wall's corner, that box's corner hit the wall while the marble itself was still a few pixels away, so it bounced off empty space.

The fix replaces `_resolve_wall_collisions` in [game/game_engine.py](#):

1. It finds the point on each wall closest to the marble's centre.
2. It only counts a hit if that point is closer than the marble's radius, so it's a real circle check.
3. It pushes the marble back out along that direction. Hitting a corner now gives a natural diagonal bounce instead of a sharp sideways one.
4. It only reverses the part of the speed heading into the wall. The marble can still slide along a wall instead of stopping dead.

The old bounce strength (0.3) is unchanged, so the game feels the same apart from the fix.

I tested it headlessly rather than by playing:

Case	Result
Marble 2px from the corner of a wall (old code bounced here)	No bounce ✓
Marble touching the underside of a wall	Pushed out and bounces down ✓
Marble hitting the corner at an angle	Diagonal bounce ✓
Marble sliding along the bottom wall	Keeps its sideways speed ✓
3,000 frames of random movement	Never got stuck inside a wall ✓

To see it yourself, press **F5** in IDLE and drag the marble along the underside of the top wall towards its right end. It should only bounce when it visibly touches the wall.

I haven't committed anything; we can push once all four tasks are done. For your chat export, you could use this as your Task 1 prompt: *"Fix the collision detection so the round marble only bounces when it actually touches a wall, not near corners where its bounding box overlaps."* Ready for Task 2, the end screen?

ME (AADARSH)

also before that can you clone this repo into a folder called lab 4 in my SE Lab repo

- Bash: Check the SE Lab repo's remote and sync status
- Bash: Copy the maze code into a lab 4 folder in the SE Lab repo

CLAUDE

I've copied the maze code into a new lab 4 folder in your local SE Lab repo. It's named to match your other lab folders:

```
Documents/Sem5/SE/Lab 1/          ← your SE Lab repo
├─ lab 4/
│   └─ 42_marble-tilt-maze/
│       ├── README.md
│       ├── main.py
│       ├── requirements.txt
│       └─ game/ (game_engine.py, marble.py, wall.py)
```

- **It's a plain copy, not a git clone.** A clone inside your SE Lab repo would bring its own git history, and GitHub would show the folder as an empty, unclickable link instead of your code. I left out the `__pycache__` files as well.
- **It already includes the Task 1 fix.**
- **Nothing is committed or pushed yet.** It shows up as a new, untracked folder.

We'll keep working in the fork at Sem5/SE/Lab 4/42_marble-tilt-maze, and I'll copy each task into lab 4 as we finish it. That way your fork and your Lab repo end up with the same code.

Your "before" video isn't on GitHub yet. You can put it directly in lab 4 (for example as lab 4/before.mov), and we'll commit it together with everything else at the end.

Shall I start Task 2, the end screen?

ME (AADARSH)

lets push the video with that name

- Bash: Copy the before video into lab 4, commit and push only that file

CLAUDE

The "before" video is pushed. It's at lab 4/before.mov in [PES1UG24CS706 Aadarsh Lab SE](#), in commit 978bcd5 ("Add Lab 4 before video").

That commit has only the video. The maze code in lab 4 and your other untracked files (.DS_Store and the two .pages files) haven't been committed. We'll commit the code once all four tasks are done.

Ready for Task 2, the end screen?

ME (AADARSH)

but i want all the files from the original repo to be in the lab 4 folder

- Bash: Commit original starter code to lab 4, push, then restore the Task 1 fix locally

CLAUDE

All the original repo files are now in lab 4 on GitHub, in commit fe07e92:

```
lab 4/
├─ before.mov
├─ 42_marble-tilt-maze/
│   ├── README.md
│   ├── main.py
│   ├── requirements.txt
│   └─ game/ (game_engine.py, marble.py, wall.py)
```

- **I pushed the untouched starter code**, not the version with the Task 1 fix, so the commit matches the original repo exactly. I left out only the `__pycache__` files, which are temporary Python files and

not real code.

- **The Task 1 fix is back in place locally** and shows as one modified file, `game_engine.py`. When we commit each task, your commit history will show the change from the original code.
- **The code sits in a 42_marble-tilt-maze subfolder.** If you'd rather have the files directly in lab 4, next to the video, I can move them.

Shall I commit and push Task 1 now, or start Task 2?

ME (AADARSH)

lets have them directly in lab 4

- Bash: Move maze files up into lab 4, commit and push the move

CLAUDE

The files now sit directly in lab 4, pushed in commit b58b4cf:

```
lab 4/  
├── before.mov  
├── README.md  
├── main.py  
├── requirements.txt  
└── game/ (game_engine.py, marble.py, wall.py)
```

The move went up as its own commit, and the files still contain the original starter code. Your Task 1 fix is still only on your laptop, as a change to `lab 4/game/game_engine.py`.

We'll keep editing in the fork at `Sem5/SE/Lab 4/42_marble-tilt-maze` and copy each task into lab 4. That's also the copy open in IDLE.

Shall I commit and push Task 1 to both repos now, or start Task 2?

ME (AADARSH)

so since we have finished task 1 ill test it on my laptop locally and then we can push the new code

CLAUDE

Good plan. Run it from the IDLE window that has `main.py` open, from the fork folder `Sem5/SE/Lab 4/42_marble-tilt-maze`, and press **F5**. If `main.py` isn't open anymore, use `File > Open`.

What to check:

1. **The corner bug is gone.** Tilt the marble up against the underside of the top wall and slide it right towards the wall's end (around the middle of the screen). It should only bounce when it visibly touches the wall.
2. **Corners feel natural.** If it clips a corner, it should glance off at an angle.
3. **Normal walls still work.** It bounces off walls, slides along them, and never passes through or gets stuck.
4. **The goal still works.** Rolling into the green circle should still print `Solved!` ... in the IDLE Shell. The proper end screen is Task 2.

If you had IDLE open before the fix, close and reopen `main.py` so you're running the new code.

Tell me how it goes. If anything looks off, describe it and I'll fix it. If it's all good, I'll push Task 1 to your fork and your Lab repo.

ME (AADARSH)

ok push it with the commit and also the repo should have language used as python

- Bash: Commit and push Task 1 to fork and Lab repo, then check GitHub languages

CLAUDE

Task 1 is pushed to both repos with the same commit message:

Repo	Commit
Fork: 42_marble-tilt-maze	39f00f6
Lab repo: PES1UG24CS706_Aadarsh_Lab_SE (Lab 4/)	6636054

Task 1: Fix collision detection with circle-vs-rectangle test Replace the square bounding-box check with a true circle-vs-rectangle test so the marble only bounces when it actually touches a wall, removing the phantom bounces near wall corners.

Language: GitHub now lists both repos as **Python**, so there was nothing to change. GitHub ignores the PDFs, images, video and the README when it works out a repo's language, so the .py files are all it counts. On the Lab repo the label can take a few minutes to appear on the page.

Running the game left some Python cache files (`__pycache__`) in the fork. I didn't commit them. Your fork still includes the ones from the original starter repo, and I can remove those and add a `.gitignore` if you want.

Ready for Task 2, the end screen?

ME (AADARSH)

ok lets go to task 2

- Read: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`

CLAUDE

Now I'll make the edits: a `reset()` so the round can restart, key handling on the end screen, a frozen timer, and the overlay.

- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Read: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/main.py`
- Read: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/main.py`
- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/main.py`

CLAUDE

Now testing both endings headlessly and saving screenshots of the end screens:

- Bash: Headlessly test win/timeout end screens and key handling
- Read: `/private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/win.png`
- Read: `/private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/lose.png`

CLAUDE

It works. One problem: a maze wall shows through behind "Finished in 12.3s", so I'll put a solid panel behind the text.

- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Bash: Re-render the win end screen with the panel
- Read: `/private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/win.png`

CLAUDE

Task 2 is done and tested headlessly. The game now shows a proper end screen instead of printing the result to the IDLE Shell.

When you roll into the goal you get a green **"Maze Solved!"** box with your finish time. When time runs out you get a red **"Time's Up!"** box. Both look like the win screen above, with the maze dimmed behind them.

What changed:

- **The game waits on the end screen.** The marble and the timer both freeze. Press **R** (or Enter or Space) to play again, or **Esc** (or Q) to quit, which closes the window cleanly.
- **The keys only work on the end screen**, so pressing Esc mid-game won't end your run by accident.
- **The code changes:** `game_engine.py` gets a `reset()` method for restarting, key handling in `handle_event`, and a new `_render_end_screen` that replaces the old print lines. `main.py` gets a two-line check so Esc can exit the loop.

What I tested headlessly:

Check	Result
Reaching the goal shows the win screen with the correct time (12.3s)	✓
Running out of time shows the "Time's Up!" screen	✓
Marble and timer freeze on the end screen	✓
R restarts with the marble back at the start	✓
Esc ignored mid-game, quits on the end screen	✓

For now R replays at the same settings. Task 3 will change it to a difficulty menu (Easy, Medium, Hard). Close and reopen `main.py` in IDLE, press **F5**, and try both a win and a time-out. Tell me if it's good and I'll push Task 2 to both repos.

ME (AADARSH)

ok push it with the commit

- Bash: Commit and push Task 2 to fork and Lab repo, verify both match

CLAUDE

Task 2 is pushed to both repos:

Repo	Commit
Fork: 42_marble-tilt-maze	49cab75
Lab repo: PES1UG24CS706_Aadarsh_Lab_SE (Lab 4/)	424413f

Task 2: Add game over screen Show an end screen when the round ends instead of printing to the console: "Maze Solved!" with the finish time, or "Time's Up!" if the timer runs out. The marble and timer freeze, and the game waits for input: R/Enter/Space to play again, Esc/Q to quit.

I compared the code in the two repos and it's identical.

Next is Task 3: after the end screen, the player chooses Easy, Medium or Hard, or exits. Ready?

ME (AADARSH)

lets move onto the next task

- Read: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py
- Edit: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py
- Edit: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py
- Edit: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py
- Edit: /Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py
- Bash: Headlessly test difficulty selection, clicks, keys and exit
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/end3.png

CLAUDE

Task 3 is done and tested headlessly. The end screen now lets you pick a difficulty for the next round or exit, as in the render above.

The difficulty levels:

	Tilt strength	Friction	Time limit
Easy	0.45 (gentler)	0.035 (slows faster)	60s
Medium	0.6	0.02	45s
Hard	0.8 (sharper)	0.01 (slidier)	30s

Medium uses the original game's values, and the first round starts on Medium.

How it works:

- **Click a button** to start a new round at that difficulty. The buttons highlight when you hover over them. The keyboard also works: **1, 2 or 3** to choose, **Esc or Q** to exit.
- **The current difficulty shows in the top-right corner** during play, in that level's colour.
- **Clicks and keys do nothing mid-game**, so you can't change level or quit by accident.
- **R no longer replays**. Replaying now means choosing a difficulty, as the README asks.
- **Only game_engine.py changed**. It has a DIFFICULTIES table at the top, a set_difficulty() method, and the buttons and click handling on the end screen. main.py is unchanged.

What I tested headlessly:

Check	Result
Starts on Medium with the original settings	✓
Clicking Easy applies 0.45 / 0.035 / 60s and restarts the marble	✓
Easy doesn't time out until 60s	✓
Pressing 3 switches to Hard	✓
Clicking Medium switches back	✓
Clicking Exit quits	✓
Clicking empty space on the end screen does nothing	✓

Check	Result
Keys and clicks during play are ignored	✓

I only tested these settings in code, not by playing. When you play, check that Hard feels hard but beatable in 30s and Easy doesn't feel too sluggish. I can adjust any of the numbers.

Close and reopen `main.py` in IDLE, then **F5**. When you're happy, tell me and I'll push Task 3.

ME (AADARSH)

i mean i was able to do it in 21 seconds but yea ig the time does make it harder

CLAUDE

Finishing Hard in 21 seconds means it's beatable but tight, which is what a Hard level should be.

If you want it harder, the simplest change is cutting Hard's time from 30s to 25s. You'd still have been 4 seconds inside it, but a messier run would lose. Stronger tilt or less friction would make the marble harder to control, but they'd make it feel slippery rather than challenging.

I'd leave it at 30s. The three levels step down evenly (60, 45, 30), and it's easy to explain in the README.

Should I push Task 3 as it is, or change Hard to 25s first?

ME (AADARSH)

change to 25s

- Edit: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/game_engine.py`
- Bash: Verify Hard time limit is now 25s

CLAUDE

Hard now has 25 seconds, and its end-screen button shows "25s". I checked that choosing Hard sets the limit to 25s.

	Tilt	Friction	Time
Easy	0.45	0.035	60s
Medium	0.6	0.02	45s
Hard	0.8	0.01	25s

Reopen `main.py` in IDLE and press **F5** to try it. Tell me when you want Task 3 pushed.

ME (AADARSH)

i just finished it in 3.4s

CLAUDE

3.4 seconds is surprisingly fast, so I'll check whether that's physically possible or a bug:

- Bash: Simulate a near-optimal autopilot run on each difficulty

CLAUDE

Your 3.4 seconds is a real result, not a bug. I had the computer steer the marble along the shortest route on each level and checked it never passed through a wall:

Level	Computer's route time	Passed through a wall?
Easy	4.2s	No
Medium	3.7s	No
Hard	3.6s	No

You beat the computer by cutting corners better. On Hard the marble reaches top speed quickly and there's little friction, so a skilled run is very fast.

The maze is short, so the time limit only catches someone who crashes around a lot. Even 25s leaves a lot of slack for a good player. What actually makes Hard hard is control: the stronger tilt and lower friction make it easy to overshoot the gaps.

That's fine for the assignment. The README asks for a difficulty choice that changes tilt strength, friction and time limit, and yours does all three. If you want Hard to be properly punishing, you could:

- **Cut the time much further**, to around 10–12s, so mistakes cost you.
- **Raise Hard's top speed**, which is 9 on every level right now, to something like 12, so it overshoots even more.

I'd leave it as it is and push. Should I push Task 3 with Hard at 25s, or change it first?

ME (AADARSH)

ok push with the commit

- Bash: Commit and push Task 3 to fork and Lab repo, verify both match

CLAUDE

Task 3 is pushed to both repos:

Repo	Commit
Fork: 42_marble-tilt-maze	743bf78
Lab repo: PES1UG24CS706_Aadarsh_Lab_SE (Lab 4/)	be1e290

Task 3: Add replay option with difficulty selection After the end screen, the player can play again by choosing Easy, Medium or Hard (click a button or press 1/2/3), or exit (Exit button or Esc/Q). Each difficulty sets tilt strength, friction and time limit: Easy 0.45 / 0.035 / 60s, Medium 0.6 / 0.02 / 45s (original settings), Hard 0.8 / 0.01 / 25s. The current difficulty is shown in the top-right corner.

The code in the two repos is identical. A macOS .DS_Store file appeared in the fork folder; I didn't commit it.

Last one is Task 4: sounds for bouncing off a wall, reaching the goal and the timer running out. Ready?

ME (AADARSH)

lets move onto task 4

- Write: `/Users/aadarsh/Documents/Sem5/SE/Lab 4/42_marble-tilt-maze/game/sounds.py`

CLAUDE

Now hooking the sounds into the engine:

- Bash: Wire sound effects into the engine and main.py

CLAUDE

Testing when each sound fires, including that a marble resting against a wall stays silent:

- Bash: Test when bounce, win and timeout sounds are triggered

CLAUDE

One more check: the game must still run if there's no audio device.

- Bash: Verify the game runs silently without an audio device

CLAUDE

Task 4 is done. The game now plays sounds, and they're generated in code, so there are no audio files to add to the repo.

Event	Sound
Bouncing off a wall	A short low thud, louder for harder hits
Reaching the goal	A rising four-note jingle, about 0.6s
Timer running out	A falling "wah-wah-wah", about 0.85s

Details:

- **No buzzing when resting on a wall.** Holding the marble against a wall pushes it into the wall every frame, which would play the thud nonstop. Only real impacts make a sound, and there's a short pause between thuds so rattling in a corner doesn't turn into a buzz.
- **It still runs without sound.** If there's no audio output, the game runs silently instead of crashing.
- **The code changes:**
 - A new file, `game/sounds.py`, builds and plays the three sounds.
 - `game_engine.py` plays them at the right moments. The wall-collision check now also reports how hard the hit was.
 - `main.py` gets one line that shortens the audio delay so sounds play instantly.

What I tested headlessly:

Check	Result
Sound works with your Mac's real audio output	✓
Marble resting against a wall for 2s on each level makes no sound	✓
A hard hit on a wall plays one thud	✓
Reaching the goal plays the jingle once	✓
Running out of time plays the timeout sound once	✓
With no audio device, the game still runs silently	✓

I couldn't hear the sounds myself, so check they sound right. Reopen `main.py` in IDLE, press **F5**, and try wall hits, a win and a time-out with your volume up. If anything is too loud, too quiet or annoying, tell me and I'll adjust it. Then I'll push Task 4.

ME (AADARSH)

there is sound push it with commit come on

- Bash: Commit and push Task 4 to fork and Lab repo, verify both match

CLAUDE

Task 4 is pushed to both repos, and all four README tasks are now done.

Repo	Commit
Fork: 42_marble-tilt-maze	81b6a8a
Lab repo: PES1UG24CS706_Aadarsh_Lab_SE (lab 4/)	3e97f3d

The code in the two repos is identical.

What's left:

1. **Record the "after" video**, about 10 seconds. Show a wall bounce with sound (screen recordings don't capture audio unless you turn on the mic under Options in the ⌘⇧5 toolbar), a win or time-out with the end screen, and choosing a difficulty.
2. **Export this chat as a PDF or doc** for the chat-history deliverable. The README also asks for a link to the chat.
3. **Add both to lab 4**, for example `after.mov` and `chat-history.pdf`. Send them over and I'll push them the same way as the before video.

Your videos are around 33 MB each. GitHub accepts that, but it will warn you about files over 50 MB, so keep the "after" video short.

ME (AADARSH)

ok i need to now make the after video right ?

CLAUDE

Yes, the "after" video is next. You have a lot to fit into about 10 seconds, so here's an order that shows all four tasks:

1. **Before recording:** reopen `main.py` in IDLE so you're running the latest code, and turn your volume up.
2. **Start the recording** with ⌘⇧5, and choose "Record Selected Portion" around the game window.
3. **Hit a wall hard (Task 4).** You should hear the thud.
4. **Slide along the underside of a wall past its corner (Task 1).** It should only bounce when it visibly touches.
5. **Roll into the goal (Task 2).** The "Maze Solved!" screen appears with your time and the jingle plays. On Medium you can finish in about 4 seconds.
6. **Click Hard (Task 3).** The new round starts, with "Hard" showing in the top corner.
7. **Stop the recording.** Stop from the menu bar, or press ⌘^Esc.

Sound: macOS screen recording doesn't capture the game's audio directly. Under **Options** in the ⌘⇧5 toolbar, set the microphone to **MacBook Pro Microphone** so it picks up the sound from your speakers. Keep the room quiet.

If it runs a bit over 10 seconds, that's fine; your before video was 23. Send me the file when it's done and I'll check it and push it as `lab 4/after.mov`.

ME (AADARSH)

@"/Users/aadarsh/Documents/Pictures/Screenshots/Screen Recording 2026-09-30 at 9.55.45 PM.mov"
ok?

- Bash: Copy the after video and extract 10 frames plus audio info
- Bash: Extract duration, audio info and 10 frames from the after video
- Bash: Combine the 10 frames into one contact sheet
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/after/sheet.jpg

CLAUDE

The video looks good. It has an audio track too, so I'll check that it actually picked up sound rather than silence:

- Bash: Measure audio loudness over time in the after video

CLAUDE

The video is mostly good, and the sound came through clearly.

What it shows:

Feature	In the video?
Task 4: sound	✅ Six wall thuds between 0.3s and 4.7s, then the win jingle at 5.9–6.7s. It's silent in between, so the game's audio was recorded directly, not picked up by the mic from your speakers.
Task 1: collisions	✅ The marble bounces cleanly through the whole maze with no phantom bounces.
Task 2: end screen	✅ "Maze Solved!" with "Finished in 6.1s", and the timer frozen at 38s.
Task 3: difficulty	⚠️ The buttons appear, but you never click one, so the video doesn't show replaying at a new difficulty.

It's 8.3 seconds long, under the 10 the handout asks for. It's also only 0.6 MB, which is fine to push.

The one gap is Task 3. The video ends on the end screen without choosing a difficulty. Since there's time left, I'd re-record the same run and add about 2 seconds at the end: click **Hard** and show the new round starting, with "Hard" and "Time: 25s" in the corners.

If you'd rather not re-record, this one is acceptable, because the difficulty buttons are clearly visible. Should I push it as it is, or will you re-record?

ME (AADARSH)

@"/Users/aadarsh/Documents/Pictures/Screenshots/Screen Recording 2026-09-30 at 10.03.18 PM.mov"

- Bash: Analyze the new after video's frames and audio
- Read: /private/tmp/claude-501/-Users-aadarsh-Documents-Internship-Monolith/293e11d7-5e9e-484d-a10b-616ed3e56487/scratchpad/after2/sheet.jpg
- Bash: Copy the after video into lab 4, commit and push it

CLAUDE

The after video is pushed as lab 4/after.mov, in commit e1dae53.

Deliverable	Status
Before video	✅ lab 4/before.mov
After video	✅ lab 4/after.mov
Updated code	✅ All four tasks, in the Lab repo's lab 4/ and in your fork
Chat history as a PDF or doc	🕒 Still to do

Last step: export this chat.

- **In the Claude desktop app:** there's no one-click PDF export here. Press ⌘P and choose "Save as PDF", or select the conversation, copy it and paste it into Pages, then export it as a PDF.
- **If you used a separate chat** (ChatGPT, Gemini or claude.ai) for the prompts you're submitting, export or print that one instead. The README also asks for its share link, which you can paste into your submission.

Save it as something like `chat-history.pdf`, send it over, and I'll push it to `lab 4`. That completes Lab 4.

ME (AADARSH)

push the video and then ill make the pdf

- Bash: Confirm the after video is on GitHub

CLAUDE

The video was already pushed in the last step, and I've just checked GitHub to confirm. `after.mov` (1.1 MB) is in your Lab repo's [lab 4 folder](#), next to `before.mov` and the code. Your local copy is in sync with GitHub.

Go ahead and make the PDF. Send me the file when it's ready and I'll push it to `lab 4`.

ME (AADARSH)

i am unable to make a pdf of this chat

- ToolSearch: ToolSearch